

MINI PROJECT

COMPLETE LOCAL WORDPRESS DEVELOPMENT SETUP

Name: Muhammad Taha

Registration Number: AUG26-WD27-20

Date: 7 September 2026

Introduction:

This mini project demonstrates the setup and configuration of a complete local WordPress development environment. The project includes custom database configuration, essential plugins, an Astra child theme, custom action and filter hooks, and a custom user role with defined capabilities.

Environment Setup and Database Configuration:

I created a local WordPress development environment using XAMPP. Apache and MySQL were started through the XAMPP Control Panel, and a separate project folder was created for the website.

A custom MySQL database was configured for the WordPress installation using phpMyAdmin. WordPress was then successfully installed and connected to the local database.

For enhanced security, the default database table prefix was changed from wp_ to taha_ during setup. Additionally, debugging was enabled in wp-config.php by setting `define('WP_DEBUG', true);` to capture errors during development.

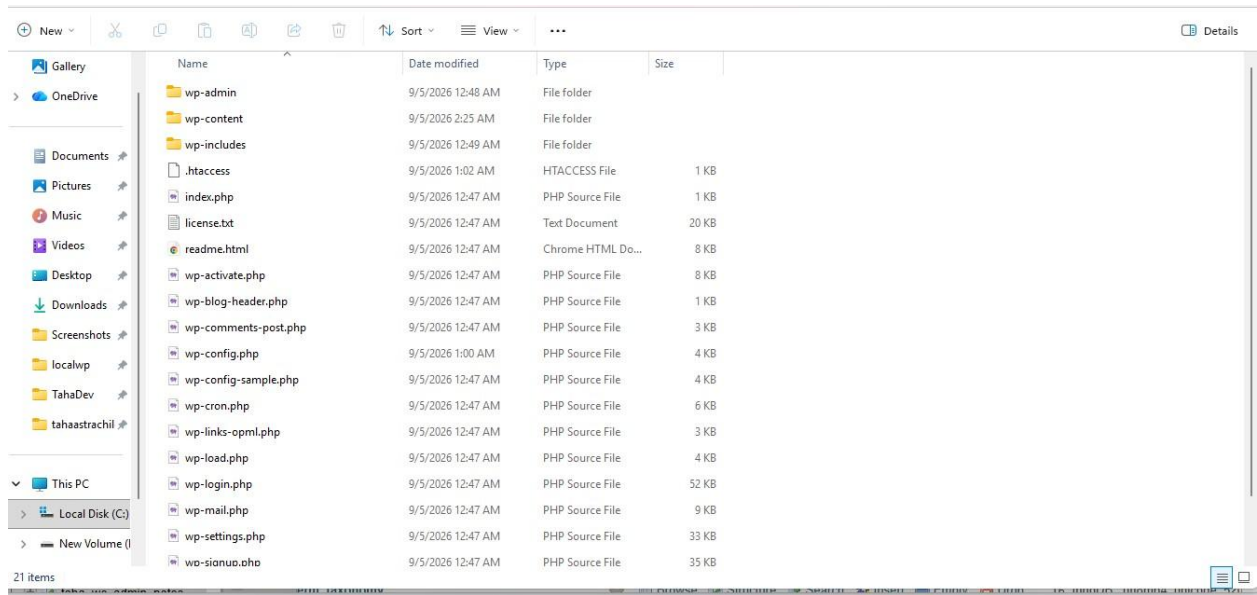


Figure 1: Wordpress File Structure.

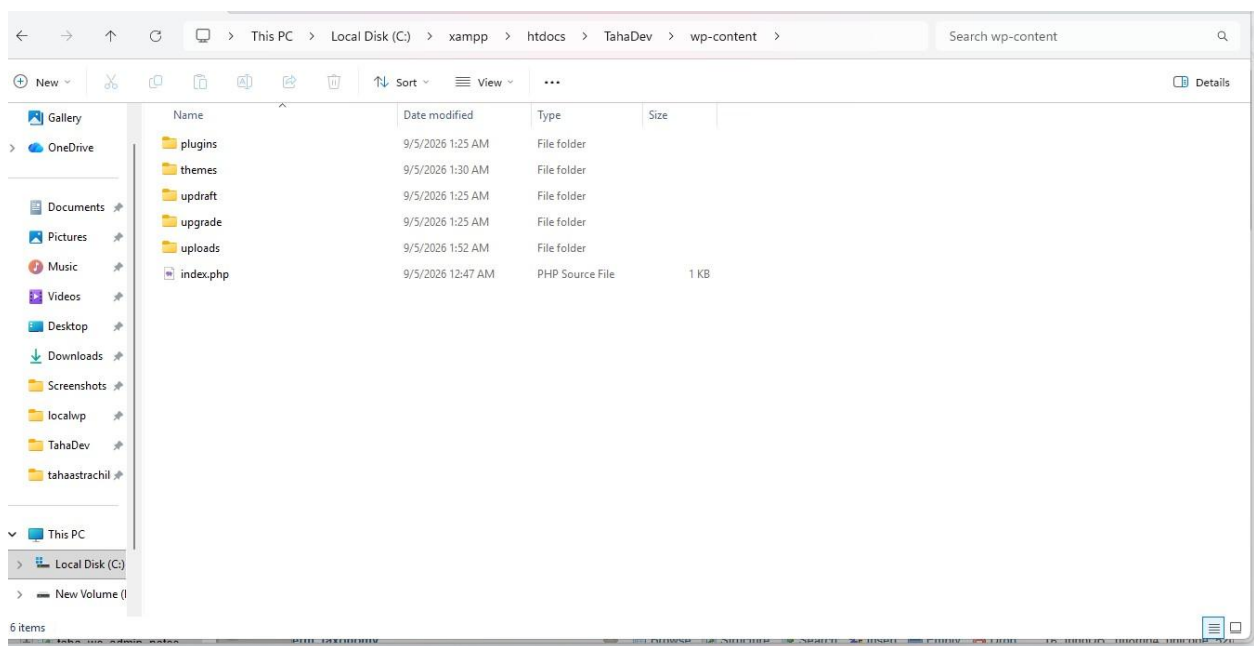


Figure 2: wp-content directory Structure.

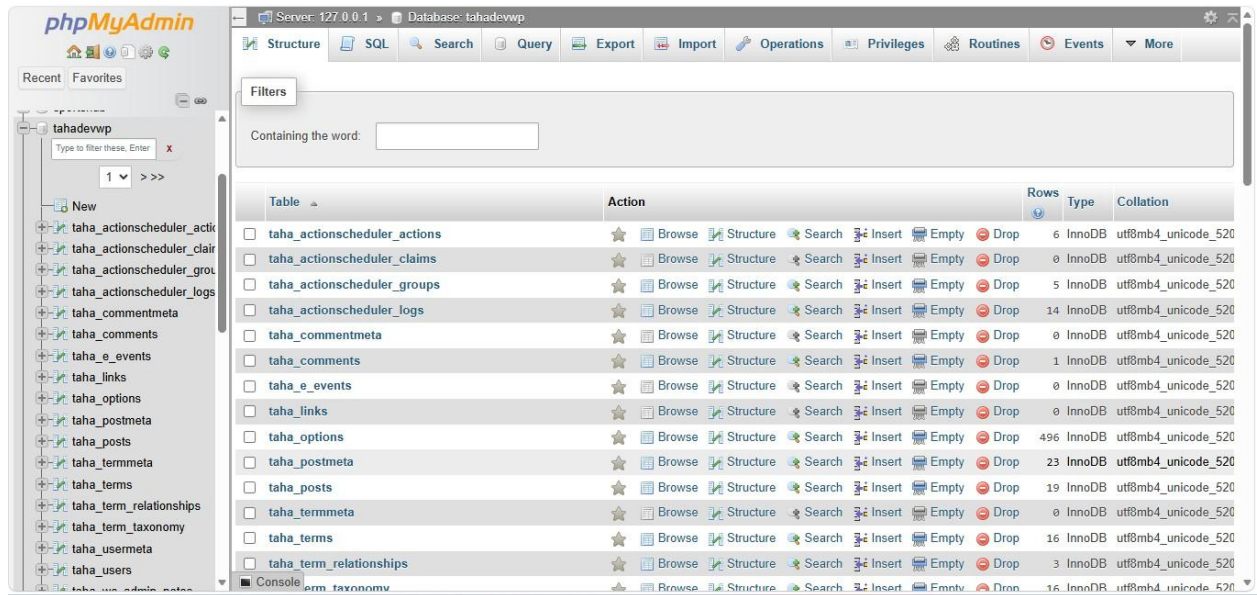


Figure 3: WordPress Database in phpMyAdmin.

Theme and Essential Plugins:

Theme:

The Astra theme was installed and activated as the base theme for the WordPress website. A custom child theme named Taha Astra Child was also created to safely add custom styles and functionality.

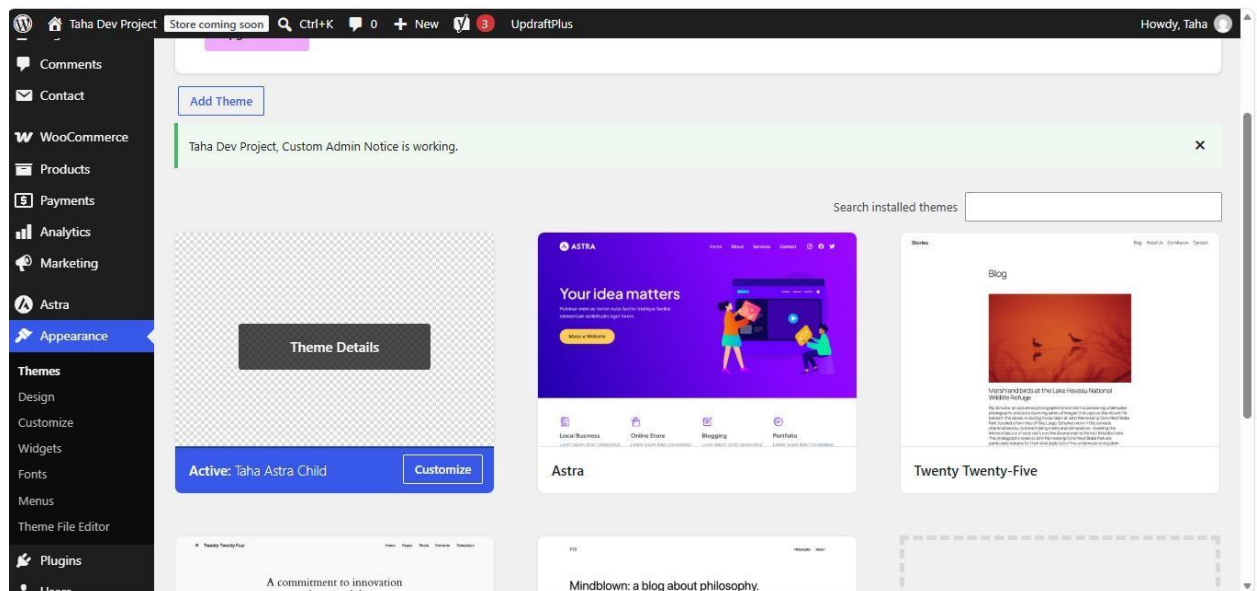


Figure 4: Activated Taha Astra Child Theme.

Essential Plugins:

The following five plugins were installed and activated:

- Elementor.
- WooCommerce.
- Contact Form 7.
- Yoast SEO.
- UpdraftPlus.

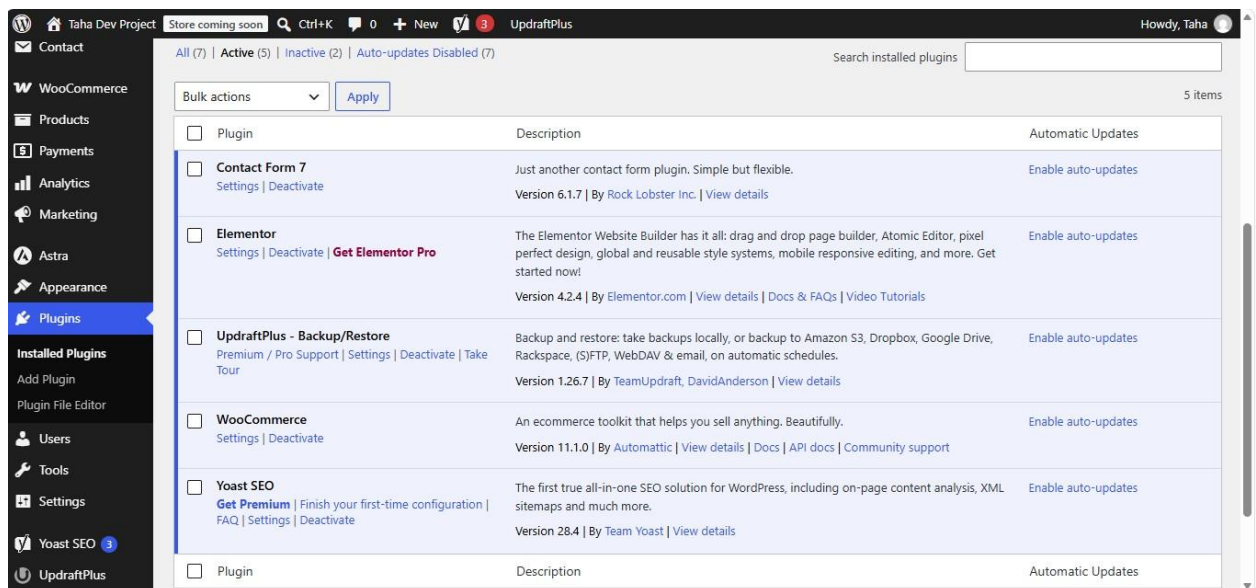


Figure 5: Installed and Activated WordPress Plugins.

Custom Action and Filter Hooks:

Custom hooks were added to the child theme's `functions.php` file to demonstrate WordPress actions and filters. These hooks modify the website output and add custom functionality.

Action Hooks:

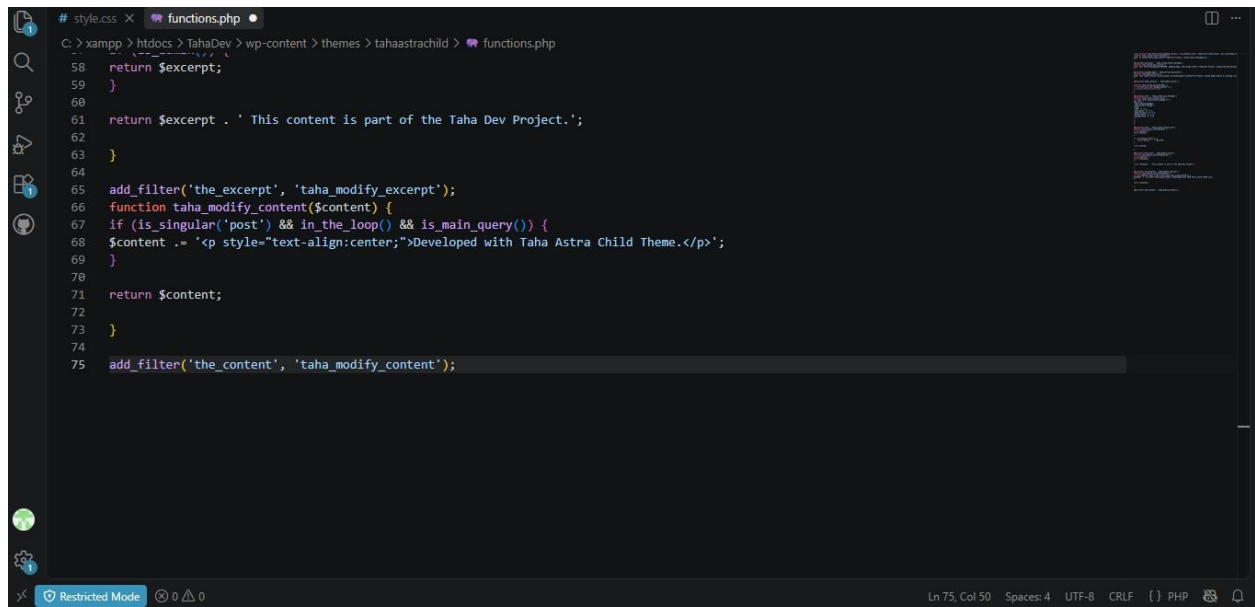
Five action hooks were implemented, including `wp_enqueue_scripts`, `wp_footer`, `wp_body_open`, `admin_notices`, and `init`.

```
# style.css X functions.php
C: > xampp > htdocs > TahaDev > wp-content > themes > tahaastrachild > functions.php
1 <?php function taha_astra_child_enqueue_styles() { wp_enqueue_style( 'taha-astra-child-style', get_stylesheet_directory_uri() . '/style.css', ar
2 function taha_custom_footer_message() {
3 echo '<p style="text-align:center;">Taha Dev Project, Custom Footer Message</p>';
4 }
5
6 add_action('wp_footer', 'taha_custom_footer_message');
7 function taha_custom_top_notice() {
8 echo '<div style="background:#f0f0f0; padding:10px; text-align:center;">Taha Dev Project, Custom Top Notice</div>';
9 }
10
11 add_action('wp_body_open', 'taha_custom_top_notice');
12 function taha_admin_notice() {
13 echo '<div class="notice notice-success is-dismissible"><p>Taha Dev Project, Custom Admin Notice is working.</p></div>';
14 }
15
16 add_action('admin_notices', 'taha_admin_notice');
17
18 function taha_custom_init_message() {
19 if (current_user_can('manage_options')) {
20 // Custom initialization hook
21 }
22 }
23
24 add_action('init', 'taha_custom_init_message');
25 function taha_create_custom_role() {
26 if (!get_role('taha_content_manager')) {
27 add_role(
28 'taha_content_manager',
29 'Taha Content Manager',
```

Figure 6: Custom Action Hooks in functions.php.

```
# style.css X functions.php
C: > xampp > htdocs > TahaDev > wp-content > themes > tahaastrachild > functions.php
30 array(
31 'read' => true,
32 'edit_posts' => true,
33 'publish_posts' => true,
34 'delete_posts' => true,
35 'upload_files' => true
36 )
37 );
38 }
39 }
40
41 add_action('init', 'taha_create_custom_role');
42 function taha_modify_title($title) {
43 if (is_admin()) {
44 return $title;
45 }
46
47 if (is_singular('post')) {
48 | return $title . ' | Taha Dev';
49 }
50
51 return $title;
52 }
53 }
54
55 add_filter('the_title', 'taha_modify_title');
56 function taha_modify_excerpt($excerpt) {
57 if (is_admin()) {
58 return $excerpt;
59 }
```

Figure 7: Custom User Role and Action Hook Code.



```
# style.css X functions.php
C: > xampp > htdocs > TahaDev > wp-content > themes > tahaastrachild > functions.php
58 return $excerpt;
59 }
60
61 return $excerpt . ' This content is part of the Taha Dev Project.';
62 }
63
64
65 add_filter('the_excerpt', 'taha_modify_excerpt');
66 function taha_modify_excerpt($content) {
67     if (is_singular('post') && in_the_loop() && is_main_query()) {
68         $content .= '<p style="text-align:center;">Developed with Taha Astra Child Theme.</p>';
69     }
70 }
71 return $content;
72 }
73 }
74
75 add_filter('the_content', 'taha_modify_content');
```

Figure 8: Custom Filter Hooks in functions.php.

Filter Hooks:

Three filter hooks were implemented to modify post titles, excerpts, and post content.



Figure 9: Filter Hook Output on Sample post.

Custom User Role and Capabilities:

A custom user role named “Taha Content Manager” was created with specific capabilities for managing WordPress content. The role was tested to verify its assigned permissions.

WordPress dashboard interface showing the 'Add New User' form. The form includes fields for Username (required), Email (required), First Name, Last Name, Website, and Password. The Password field shows a generated password: !S^tCxc1VChUD(B9K!rC)R0Z, with a 'Hide' button. A 'Generate password' button is present. Below the password field, there is a checkbox for 'Send User Notification' which is checked, with the text 'Send the new user an email about their account'. The 'Role' dropdown is set to 'Taha Content Manager'. The left sidebar shows the 'Users' menu item highlighted.

Figure 10: Custom User Role.

WordPress dashboard interface showing the 'Edit with Elementor' editor for a new post titled 'My First Post'. The main editor area shows the title 'My First Post' and a prompt 'Type / to choose a block'. The right sidebar contains post settings including 'Set featured image', 'Generate with Elementor AI', 'Show featured image in the posts lists only, but hide it in the single post view.', 'Add an excerpt...', 'Last edited a second ago', 'Status: Draft', 'Publish: Immediately', 'Slug: my-first-post', 'Template: Default template', and 'Discussion: Open'. The bottom of the screen shows 'Meta Boxes' and 'Post' sections.

Figure 11: Post Publishing permission.

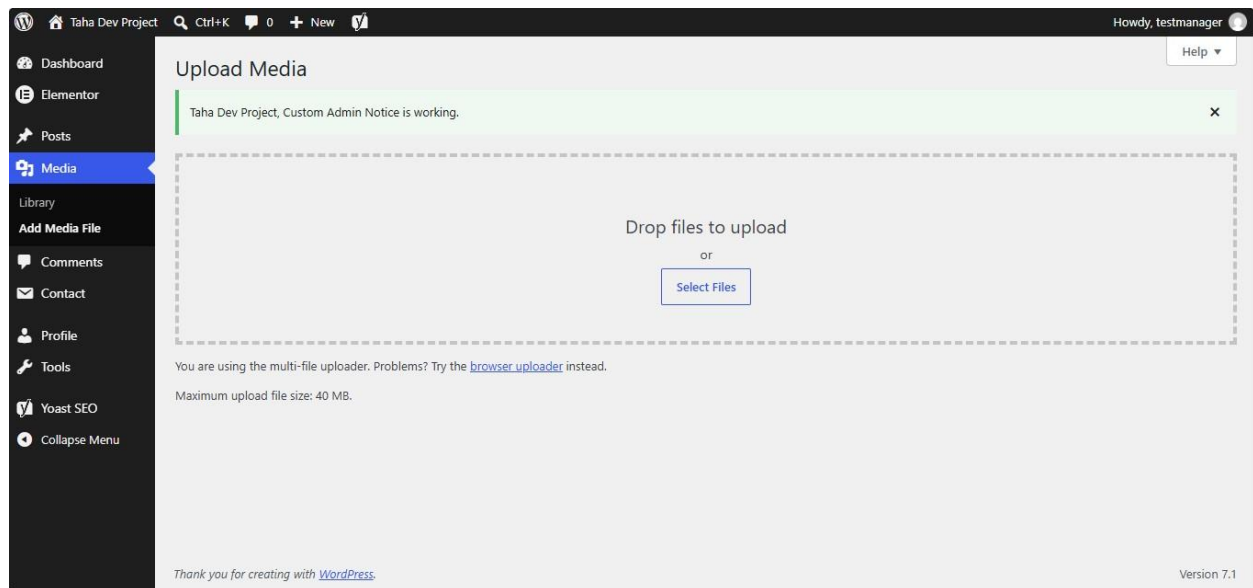


Figure 12: Media Upload permission.

Conclusion:

The local WordPress development environment was successfully configured and tested. The project demonstrates the use of a custom child theme, essential plugins, action and filter hooks, and a custom user role with defined capabilities. This setup provides a practical foundation for local WordPress development and customization.